

On Instabilities of the Taylor-Green Vortex in 2D Euler Flows

Xinyu Zhao^{1,2}, Bartosz Protas² and Roman Shvydkoy³

¹Department of Mathematics & Statistics
McMaster University, Hamilton, Ontario, Canada

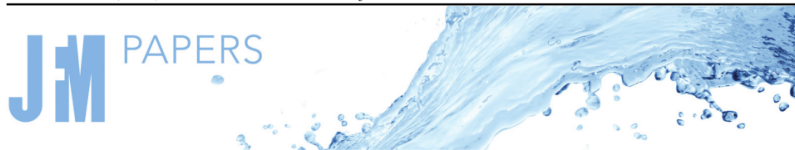
²Department of Mathematical Sciences, New Jersey Institute of Technology,
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³Department of Mathematics, Statistics and Computer Science,
University of Illinois at Chicago, Chicago, IL 60607, USA

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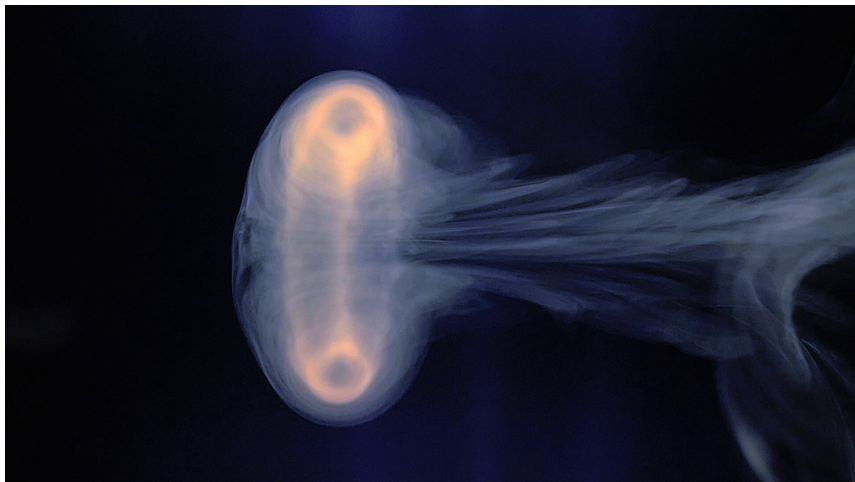
¹Department of Mathematics and Statistics, McMaster University, Hamilton, ON L8S 4K1, Canada

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Vortex Rings in Nature and Technology



Agenda

Introduction

- Taylor-Green Vortex
- Linearized Euler Operator
- Spectrum

Modal Growth of Perturbations

- Unstable Eigenvalues and Eigenvectors
- Exponential Growth
- Nonlinear Instability

Nonmodal Growth of Perturbations

- Optimization Formulation
- Flows Saturating the Spectral Bound

► 2D Euler system on a periodic domain

$$\begin{aligned}\partial_t \omega + (\mathbf{u} \cdot \nabla) \omega &= 0, & (\mathbf{x}, t) &\in \mathbb{T}^2 \times (0, T], \\ \omega|_{t=0} &= \omega_0, & \mathbf{x} &\in \mathbb{T}^2,\end{aligned}$$

where

► $\omega = \nabla \times \mathbf{u}$ is the scalar vorticity field satisfying $\int_{\mathbb{T}^2} \omega \, d\mathbf{x} = 0$

► Biot-Savart law: $\mathbf{u} = \nabla^\perp \Delta^{-1} \omega$, $\nabla^\perp = (-\partial_y, \partial_x)$

► An equilibrium solution — the 2D Taylor-Green vortex:

$$\mathbf{u}_s = (-\cos(x_1) \sin(x_2), \sin(x_1) \cos(x_2))$$

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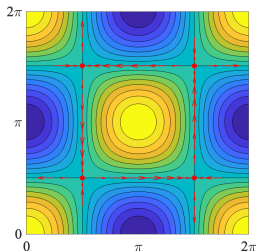
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- Linearize the 2D Euler system around a steady solution $(\omega_s, \mathbf{u}_s) \implies$

$$\begin{aligned}\partial_t w &= \mathcal{L}w, & w|_{t=0} &= w_0, & \text{where} \\ \mathcal{L}w &:= -(\mathbf{u}_s \cdot \nabla)w - (\mathbf{u} \cdot \nabla)\omega_s \\ &= \left[-\mathbf{u}_s \cdot \nabla - \nabla\omega_s \cdot (\nabla^\perp \Delta^{-2}) \right] w, & w &\in H_0^m(\mathbb{T}^2)\end{aligned}$$

- Linear stability of the equilibria determined by the spectrum of the operator $\mathcal{L}$

$$\begin{aligned}\sigma(\mathcal{L}) &:= \{\lambda \in \mathbb{C} : (\mathcal{L} - \lambda) \text{ is not bounded from below in } \mathcal{X}\} \\ &= \sigma_{\text{disc}}(\mathcal{L}) \cup \sigma_{\text{ess}}(\mathcal{L})\end{aligned}$$

- A point $z \in \sigma_{\text{disc}}(\mathcal{L})$ if
- z is an isolated point in $\sigma(\mathcal{L})$
 - z has finite multiplicity, i.e. $\dim \ker(z - \mathcal{L}) < \infty$ (finite-dimensional)
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- For the 2D linear Euler operator $\mathcal{L}$ the essential spectrum $\sigma_{\text{ess}}(\mathcal{L})$ is well understood (Friedlander et al. 1997, Shvydkoy & Latushkin 2003)

$$\sigma_{\text{ess}}(\mathcal{L}; H_0^m) = \{z \in \mathbb{C} : |\Re(z)| \leq |m| \mu_{\max}\},$$

where

$$\mu_{\max} := \lim_{t \rightarrow \infty} \frac{1}{t} \log \sup_{\mathbf{x} \in \mathbb{T}^2} \|\nabla \varphi_t(\mathbf{x})\|$$

is the maximal Lyapunov exponent corresponding to the Lagrangian flow $\varphi_t : \xi \rightarrow \mathbf{x}(t; \xi)$ generated by the steady state via $\partial_t \mathbf{x}(t) = \mathbf{u}_s(\mathbf{x}(t))$

- If $z \in \sigma_{\text{ess}}(\mathcal{L})$, then there exists a sequence of weak-null unit vectors $\{f_n\} \in H_0^m(\mathbb{T}^2)$, **approximate eigenvectors**, such that

$$\|(\mathcal{L} - z)f_n\|_{H^m} \rightarrow 0, \quad n \rightarrow \infty$$

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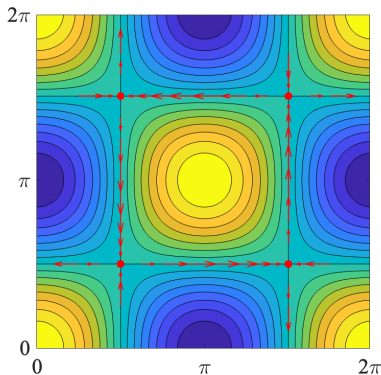
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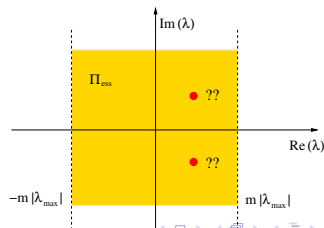
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- ▶ The problem remains open for the Taylor-Green vortex
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Theorem (Lin 2004)

Suppose there exists an exponentially growing solution $e^{\lambda t} w_0$ of the linearized system $\partial_t w = \mathcal{L}w$ with $\Re(\lambda) > 0$ and let $w_0 \in L^2(\mathbb{T}^2)$. Then we have the following

- (i) [regularity of growing modes] $w_0 \in W^{1,p}(\mathbb{T}^2) \cap L^q(\mathbb{T}^2)$ for all $1 \leq p < p^*$ and $1 \leq q < \infty$, where

$$p^* = \begin{cases} \frac{\mu_{\max}}{\mu_{\max} - \Re(\lambda)} = \frac{1}{1 - \Re(\lambda)}, & \mu_{\max} > \Re(\lambda) \\ \infty, & \mu_{\max} \leq \Re(\lambda) \end{cases}$$

- (ii) [nonlinear instability] if there exists $\epsilon > 0$, such that for any $\delta > 0$ there is a solution $w^\delta(t)$ of the 2D Euler system corresponding to the initial condition ω_0^δ , such that

$$\|\omega_0^\delta - \omega_s\|_{L^q} + \|\omega_0^\delta - \omega_s\|_{W^{1,p}} \leq \delta,$$

and

$$\sup_{0 < t < T_\delta} \|\omega^\delta(t) - \omega_s\|_{H^m} \geq \epsilon,$$

where $T_\delta = O(|\ln(\delta)|)$, $q \in [1, \infty)$, $p = [1, p^*)$ and $m \geq -1$.

- ▶ Numerical solution of the eigenvalue problem $\mathcal{L}\phi(\mathbf{x}) = \lambda\phi(\mathbf{x})$
- ▶ Discretize $\mathcal{L}$ using the basis

$$\varphi_{j_1, j_2} = \frac{1}{\sqrt{2\pi}} \cos(j_1 x + j_2 y), \quad \psi_{j_1, j_2} = -\frac{1}{\sqrt{2\pi}} \sin(j_1 x + j_2 y),$$

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to obtain a sparse algebraic eigenvalue problem $\mathbf{L}\phi = \lambda\phi$

- ▶ The spectrum of $\mathbf{L}$

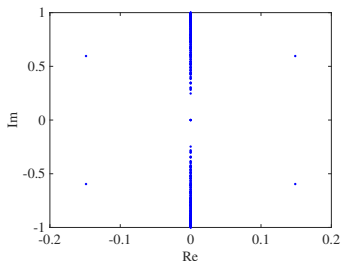
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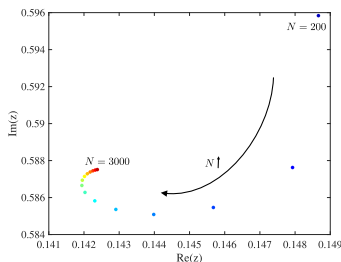
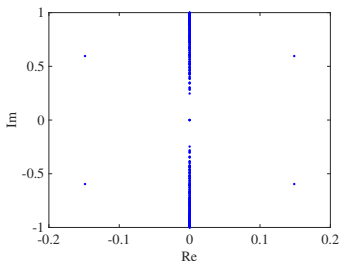
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- ▶ The unstable eigenvalue: $\lambda_+^{3000} = 0.1424 + 0.5875i$
- ▶ The corresponding eigenvectors $\Re[\phi_+^N]$ are L^2 integrable

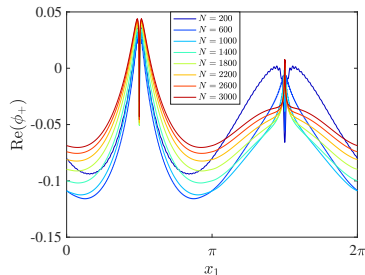
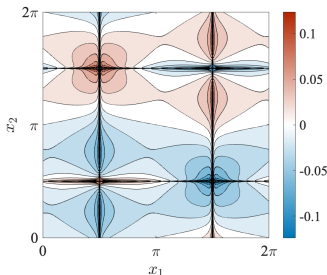
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$$\phi_+ \in W^{1,p^*}(\mathbb{T}^2), \quad \text{where} \quad p^* = \frac{1}{1 - 0.14} = 1.16$$

- ▶ From the Sobolev embedding $W^{1,p^*} \hookrightarrow H^s$ with $1/p^* - 1/2 = 1/2 - s/2$, we have

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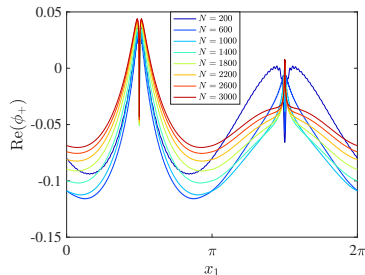
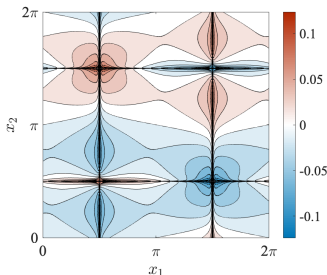
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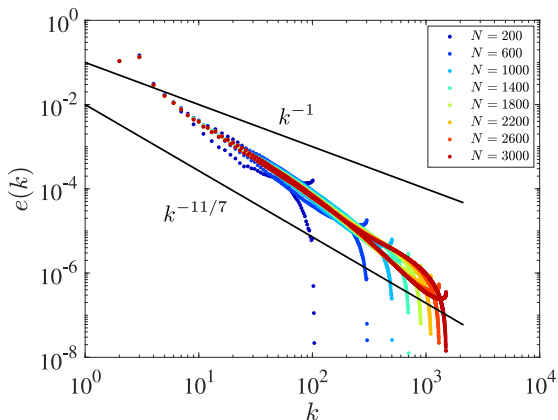
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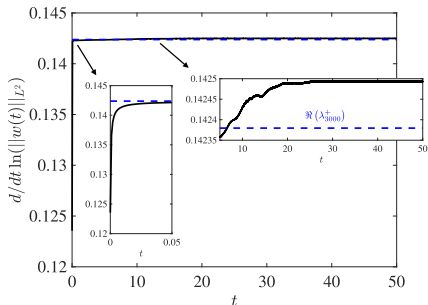
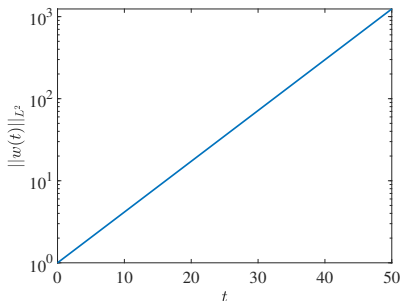
The Fourier spectrum of the eigenfunction ϕ

$$e(k) := \frac{1}{2} \sum_{k \leq |j| < k+1} |\hat{\phi}_j|^2, \quad k \in \mathbb{N}$$

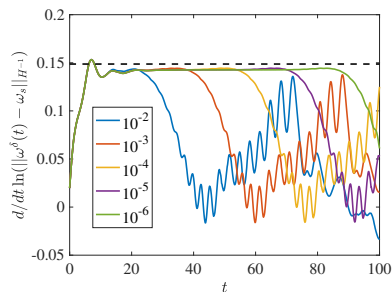
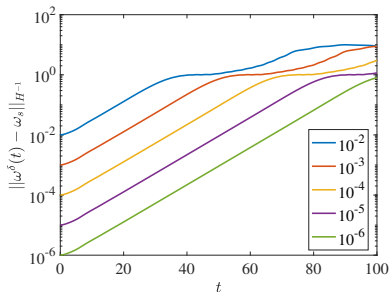


$e(k) = \mathcal{O}(k^{-1})$ implies L^2 summability; $e(k) = \mathcal{O}(k^{-1.65})$ means
 $\phi_+ \in H^{0.28}(\mathbb{T}^2)$

Exponential growth of the perturbation



The actual growth rate is within 0.01% equal $\Re(\lambda_+^{3000})$



The time-dependence of (left) the norm $\|\omega^\delta(t) - \omega_s\|_{H^{-1}}$ and (right) the growth rate $(d/dt) \ln(\|\omega^\delta(t) - \omega_s\|_{H^{-1}})$ in the solution of the nonlinear problem with the initial condition given in terms of the eigenfunction ϕ_+^{200} as $\omega_0 = \omega_s + \delta \cdot \Re(\phi_+^{200}) / \|\Re(\phi_+^{200})\|_{H^{-1}}$ with different indicated magnitudes δ .

- ▶ The growth rate of the eigenfunction with $\Re(\lambda_+0) \approx 0.1424$ does not saturate the bound on the growth of the semigroup

$$\|e^{\mathcal{L}t} w_0\|_{H^m} \leq \|w_0\|_{H^m} e^{\mu_{\max} t}, \quad \mu_{\max} = 1, \quad m = \pm 1$$

- ▶ How to find initial data w_0 realizing this growth?
- ▶ Given $T > 0$, define the objective functional $J_m : H_0^m(\mathbb{T}^2) \rightarrow \mathbb{R}$

$$J_m(w_0) = \left\| e^{T\mathcal{L}} w_0 \right\|_{H^m}^2, \quad m = \pm 1$$

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$$\max_{w_0 \in \mathcal{M}} J_m(w_0), \quad \text{where } m = \pm 1 \text{ and}$$

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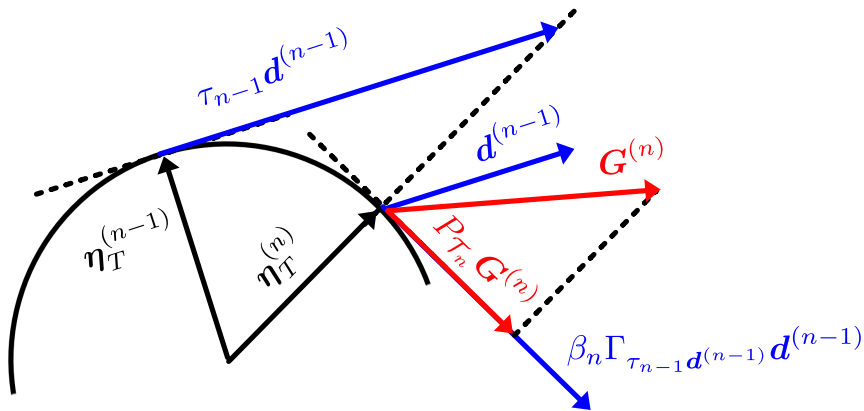
where

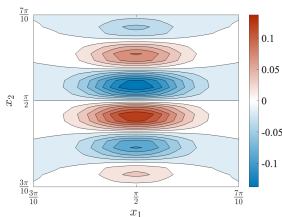
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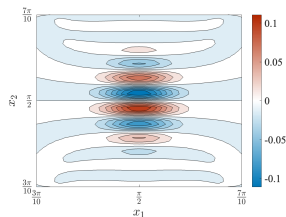
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Riemannian Conjugate-Gradient Method (Absil et al., 2008)

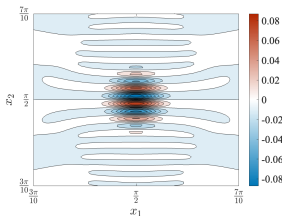




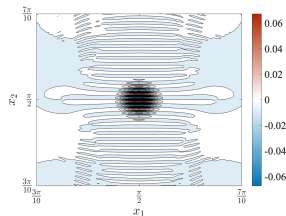
(a) $N = 128$



(b) $N = 256$

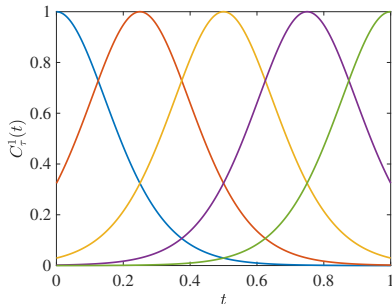
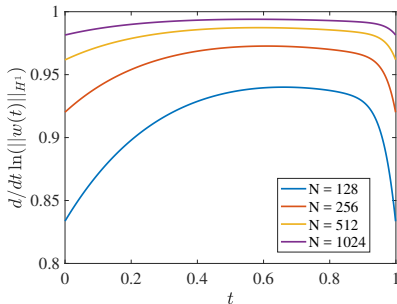


(c) $N = 512$



(d) $N = 1024$

$[m = 1]$ The optimal initial conditions $\tilde{w}_0^N$ obtained using different resolutions N .

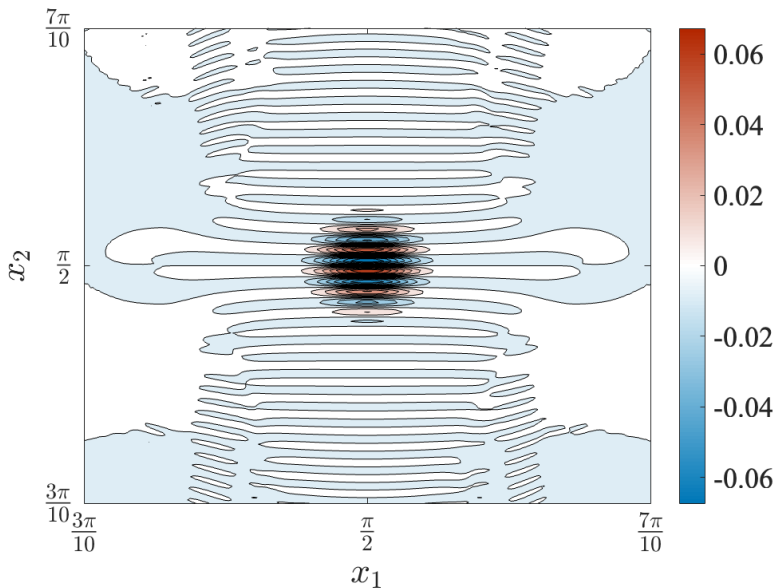


[$m = 1$] (left) The growth rate $(d/dt) \ln(\|w^N(t)\|_{H^1})$ versus t for the optimal initial conditions $\tilde{w}_0^N$ obtained using increasing spatial resolutions N^2 .
(right) The time-dependence of the autocorrelation function

$$C_\tau^1(t) := \frac{\langle w(\tau), w(t) \rangle_{H^1}}{\|w(\tau)\|_{H^1} \|w(t)\|_{H^1}}, \quad t, \tau \geq 0$$

corresponding to $N = 1024$ and $\tau = 0, 0.25, 0.5, 0.75, 1$.

Similar results obtained for $m = -1$.



Conclusions

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$$\lambda_+ \in \sigma_{\text{ess}}(\mathcal{L})$$
 - ▶ the corresponding eigenfunction given by a distribution with the Sobolev regularity consistent with Lin (2004)
 - ▶ the equilibrium also exhibits a nonmodal growth mechanism involving a continuous family of uncorrelated functions which saturates the spectral bound
 - ▶ intrinsically related to the ∞ -dimensional nature of the operator $\mathcal{L}$
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- ▶ OPEN QUESTION: stability of equilibria in 3D

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